

WASP-10b

R-0713

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_231007 · created 2026-07-27T23:08:56+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460224.7557 +/- 0.0061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.062
Transit depth (Rp/Rs)^2	3.61 +/- 2.35 [%]
Orbital Inclination (inc)	84.71 +/- 1.91
Airmass coefficient 1 (a1)	0.63 +/- 0.32
Airmass coefficient 2 (a2)	0.47 +/- 1.26
Scatter in the residuals of the lightcurve fit is	52.74 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	3.0
Transit Duration (day)	0.041 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.062 vs 0.1592 (deviation 0.5σ)
Duration fit vs published	0.041 d vs 0.0946 d (deviation 1.53σ)
Mid-time O–C	-10.4 min
Depth SNR	3.1
Residual scatter	52.74 %

Light curve

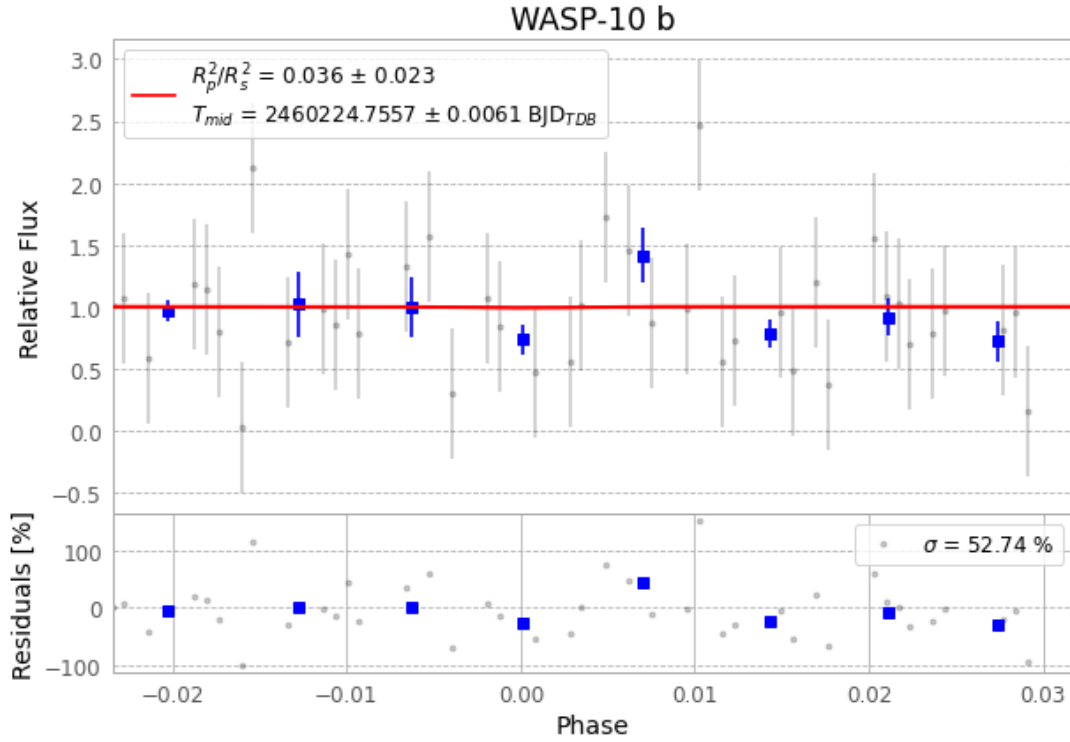


Figure 1 — FinalLightCurve_WASP-10 b_2023-10-06.png (SHA-256 e4f65e65438e73dd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [160, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 59eb568564898c2a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8a6a059

Data provenance

83 FITS frames (55 MB), WASP-10231007041515.FITS → WASP-10231007082115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-231007.log (1dff34e43706...), FinalLightCurve_WASP-10 b_2023-10-06.png (e4f65e65438e...), FinalLightCurve_WASP-10 b_2023-10-06.csv (1eeb5d97a9fa...)

Compute resources

Wall time 170.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 23:08Z.